Hw 4: Microelectronics

1) Pierret Problem 3.22

a) $p(-\infty) = p_0 = n_i^2 / N_D = 10^{20}\,cm^{-3} / 10^{18}\,cm^{-3} = 10^2\,cm^{-3}$

b) B/c of diffusion, there is a gradient

↳ $G_L = \Delta p_n(\infty) / \tau_p \Rightarrow \Delta p_n(\infty) = G_L \tau_p$

↳ $\therefore \Delta p_n(\infty) = (10^{15})(10^{-6}) = 10^9\,cm^{-3}$.

So $p(\infty) = p_0 + \Delta p_n(\infty) \simeq \Delta p_n(\infty) = 10^9\,cm^{-3}$

c) Yes b/c Δp_n will occur at $x = \infty$ (the max)
and $\Delta p_n(\infty) = 10^9\,cm^{-3} \ll n_0 = N_D = 10^{18}\,cm^{-3}$

d) $\Delta p_n(x)$ for all x

$x<0$

↳ $D_p \frac{d^2 \Delta p_n}{dx^2} - \frac{\Delta p_n}{\tau_p}$ & $\Delta p_n(x) = Ae^{-x/L_p} + Be^{x/L_p}$

↳ this is equal to 0

$\Delta p_n(-\infty) = 0 \quad \therefore A = 0$

$\Delta p_n(x) = Be^{x/L_p}$

$\therefore \frac{d\Delta p_n(x)}{dx} = \frac{B}{L_p} e^{x/L_p}$

$x>0$

$0 = \frac{D_p d^2 \Delta p_n}{dx^2} - \frac{\Delta p_n}{\tau_p} + G_L$

↳ concentration gradient

$\Delta p_n(x) = G_L \tau_p + A' e^{-x/L_p} + B' e^{x/L_p}$

↳ for $x = \infty$: $\Delta p_n(x) = G_L \tau_p \quad \therefore B' = 0$

$\therefore \Delta p_n(x) = G_L \tau_p + A' e^{-x/L_p} \rightarrow \frac{d\Delta p_n(x)}{dx} = -\frac{A'}{L_p} e^{-x/L_p}$

for $x = 0$: $B = G_L \tau_p + A'$ $\rightarrow$ $B/L_p = -A'/L_p$

$$\therefore \Delta p_n(x) = \begin{cases} \frac{1}{2} G_L \tau_p e^{x/L_p} & \text{for } x \leq 0 \\ G_L \tau_p \left(1 - \frac{1}{2} e^{-x/L_p}\right) & \text{for } x \geq 0 \end{cases}$$

2) Pierret 5.5 $\rightarrow$ $N_A = 10^{17}/cm^3$: p-side doping

$N_D = 10^{15}/cm^3$: n-side doping

a) $V_{bi} = \frac{kT}{q} \ln\left(\frac{N_A N_D}{n_i^2}\right) = 0.026 \ln\left(\frac{10^{32}/cm^6}{10^{20}/cm^6}\right) = 0.716\,V$

b) $x_p = \sqrt{\left(\frac{2K_s\varepsilon_0}{q}\right)\left(\frac{N_D}{N_A(N_A+N_D)} V_{bi}\right)} = 9.62 \times 10^{-7}\,cm$

$x_n = \sqrt{\left(\frac{2K_s\varepsilon_0}{q}\right)\left(\frac{N_A}{N_D(N_A+N_D)} V_{bi}\right)} = 9.62 \times 10^{-5}\,cm$

$W = (9.62 \times 10^{-7}\,cm) + (9.62 \times 10^{-5}\,cm) = 9.72 \times 10^{-5}\,cm$

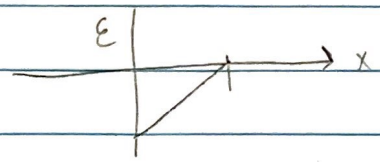
c) $\mathcal{E}(0) = \frac{-qN_D}{K_s\varepsilon_0} x_n$

$\hookrightarrow = -1.47 \times 10^4\,V/cm$

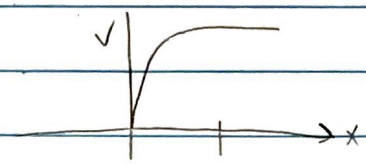
d) $V(0) = \frac{qN_A}{2K_s\varepsilon_0} x_p^2 = 7.09 \times 10^{-3}\,V$

e) ρ:

} charge density



} electric field



} electrostatic potential

3) Pierret 6.10

a) It is reverse-biased b/c at the depletion region, there is a difference between the minority carriers

b) for $x \leq -x_p$: $n_{p0} << p_p$
$\uparrow \approx \Delta n_{p\,max}$

for $x \geq x_n$: $p_{n0} << n_n$
$\uparrow \approx \Delta p_{n\,max}$

} ∴ low-level injection does prevail!

c) From above : $N_A \cong n_{p0} << p_p = 10^{14}/cm^3$

$N_D \simeq p_{n0} << n_n = 10^{15}/cm^3$

d) $V_A = \frac{kT}{q} \ln\left(\frac{n(-x_p)\,p(-x_p)}{n_i^2}\right)$ } where $n(-x_p) \approx 10^5/cm^3$, $p(-x_p) \approx 10^{14}/cm^3$

↳ $V_A = 0.026 \ln\left(\frac{10^{17}/cm^6}{10^{20}/cm^6}\right) = -0.18V$

4) Pierret 6.13 : $N_D = 10^{15}/cm^3$: n-side doping

a) $V_{Br\,approx} = 320\,V$

b) $W = \sqrt{\left(\frac{2K_s\varepsilon_0}{qN_D}\right)(V_b - V_a)}$ where $V_a = 0$

$\therefore W \approx 2.04 \times 10^{-3}\ cm$

c) $|\mathcal{E}|_{max} = \frac{qN_D}{K_s\varepsilon_0} x_n = \frac{qN_D}{K_s\varepsilon_0} W$

$\therefore |\mathcal{E}|_{max} \approx 3.13 \times 10^{5}\ V/cm$

5) PN - Junction

a) $C_j = \frac{K_s\varepsilon_0 A}{W(V_A)}$ $\therefore C_j = 1.3\,pF = \left(\frac{K_s\varepsilon_0}{W(V_A = -0.05V)}\right)(10^{-9}\,m^2)$

assymetrical : $\frac{1}{C_j^2} = \frac{2}{qK_s\varepsilon_0 A^2 N_B}(V_{bi} - V_A)$ $\therefore N_B = \frac{2(V_{bi} - V_a)}{q\varepsilon_0 A^2} C_j^2$

$\therefore N_B$ (lightly-doped) $\approx 2.04 \times 10^{17}\,cm^{-3}$

b) $V_{bi} = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right) \rightarrow 0.95V = 0.26 \ln\left(\frac{N_A N_D}{10^{20}/cm^6}\right)$

$\therefore N_A N_D = 8.7 \times 10^{35}/cm^6$

$N_{heavy} = \frac{N_A N_D}{N_{light}} \Rightarrow 4.3 \times 10^{18}/cm^3$